```
import numpy as np
import pandas as pd
```

Load the dataset

```
df = pd.read_csv('Titanic-Dataset.csv')
```

```
print(df.head())
```

```
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                             Heikkinen, Miss. Laina  female  26.0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S
```

```
print(df.tail())
```

```
     PassengerId  Survived  Pclass                                      Name  \
886          887         0       2                     Montvila, Rev. Juozas
887          888         1       1              Graham, Miss. Margaret Edith
888          889         0       3  Johnston, Miss. Catherine Helen "Carrie"
889          890         1       1                     Behr, Mr. Karl Howell
890          891         0       3                       Dooley, Mr. Patrick

        Sex   Age  SibSp  Parch      Ticket   Fare Cabin Embarked
886    male  27.0      0      0      211536  13.00   NaN        S
887  female  19.0      0      0      112053  30.00   B42        S
888  female   NaN      1      2  W./C. 6607  23.45   NaN        S
889    male  26.0      0      0      111369  30.00  C148        C
890    male  32.0      0      0      370376   7.75   NaN        Q
```

```
df.shape
```

```
(891, 12)
```

```
print(df.columns)
```

```
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
       'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='object')
```

Data Cleaning

```
print(df.isnull().sum())
```

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

```
df['Age'].fillna(df['Age'].mean(), inplace = True)
```

```
/tmp/ipykernel_18735/2940318962.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are
```

```
For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]


  df['Age'].fillna(df['Age'].mean(), inplace = True)
```

```
print(df.isnull().sum())
```

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age              0
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

```
df.dropna(inplace=True)
```

```
print(df.groupby('Sex')['Survived'].mean())
```

```
Sex
female    0.936842
male      0.420561
Name: Survived, dtype: float64
```

```
df_sorted = df.sort_values(by = 'Age', ascending=True)
print(df_sorted.head())
```

```
     PassengerId  Survived  Pclass                            Name     Sex  \
305          306         1       1  Allison, Master. Hudson Trevor    male
183          184         1       2       Becker, Master. Richard F    male
205          206         0       3      Strom, Miss. Telma Matilda  female
297          298         0       1    Allison, Miss. Helen Loraine  female
340          341         1       2  Navratil, Master. Edmond Roger    male

      Age  SibSp  Parch  Ticket      Fare    Cabin Embarked
305  0.92      1      2  113781  151.5500  C22 C26        S
183  1.00      2      1  230136   39.0000       F4        S
205  2.00      0      1  347054   10.4625       G6        S
297  2.00      1      2  113781  151.5500  C22 C26        S
340  2.00      1      1  230080   26.0000       F2        S
```

```
df['Family Size'] = df['SibSp'] + df['Parch']
print(df)
```

```
     PassengerId  Survived  Pclass  \
1              2         1       1
3              4         1       1
6              7         0       1
10            11         1       3
11            12         1       1
..           ...       ...     ...
871          872         1       1
872          873         0       1
879          880         1       1
887          888         1       1
889          890         1       1

                                                Name     Sex   Age  SibSp  \
1    Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
3         Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
6                              McCarthy, Mr. Timothy J    male  54.0      0
10                     Sandstrom, Miss. Marguerite Rut  female   4.0      1
11                            Bonnell, Miss. Elizabeth  female  58.0      0
..                                                ...     ...   ...    ...
871   Beckwith, Mrs. Richard Leonard (Sallie Monypeny)  female  47.0      1
872                           Carlsson, Mr. Frans Olof    male  33.0      0
879      Potter, Mrs. Thomas Jr (Lily Alexenia Wilson)  female  56.0      0
887                       Graham, Miss. Margaret Edith  female  19.0      0
889                              Behr, Mr. Karl Howell    male  26.0      0

     Parch    Ticket     Fare        Cabin Embarked  Family Size
1        0  PC 17599  71.2833          C85        C            1
3        0    113803  53.1000         C123        S            1
6        0     17463  51.8625          E46        S            0
10       1   PP 9549  16.7000           G6        S            2
11       0    113783  26.5500         C103        S            0
..     ...       ...      ...          ...      ...          ...
871      1     11751  52.5542          D35        S            2
872      0       695   5.0000  B51 B53 B55        S            0
```

```
879      1     11767  83.1583          C50        C            1
887      0    112053  30.0000          B42        S            0
889      0    111369  30.0000         C148        C            0

[202 rows x 13 columns]
```

```
print(df['Fare'].max())
```

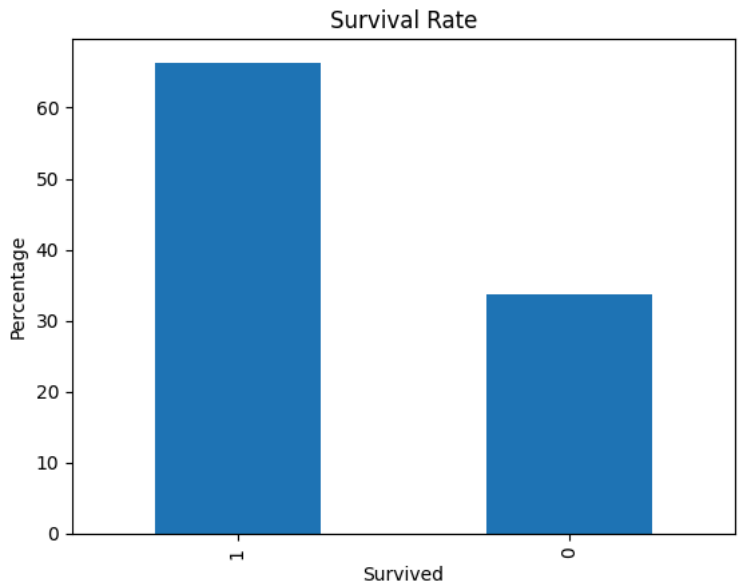
```
512.3292
```

```
df.to_csv('Cleaned Dataset.csv')
```

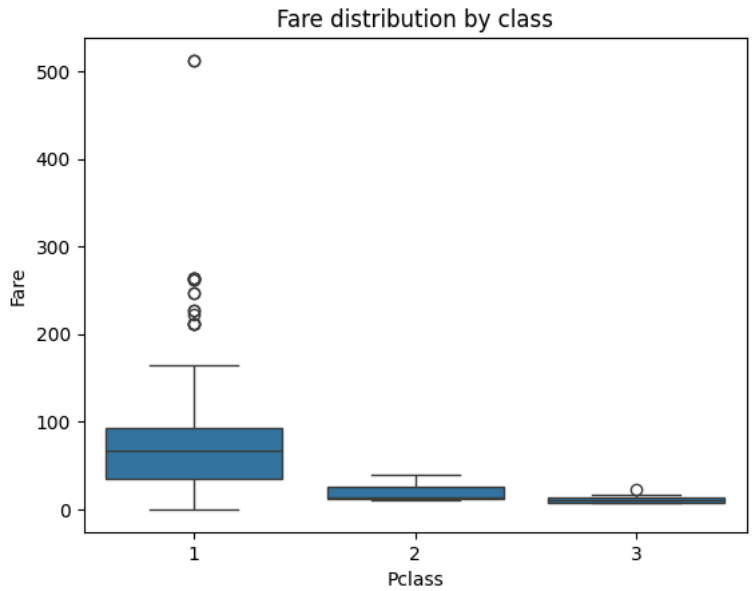
Data Visualization

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
survival_rate = df['Survived'].value_counts(normalize=True)*100
survival_rate.plot(kind='bar')
plt.xlabel('Survived')
plt.ylabel('Percentage')
plt.title('Survival Rate')
plt.show()
```




```
sns.boxplot(x = 'Pclass', y = 'Fare',data = df)
plt.title('Fare distribution by class')
plt.show()
```



```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix
```

```
df['Embarked'].fillna('S',inplace=True)
```

```
/tmp/ipykernel_18735/3895404187.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]


  df['Embarked'].fillna('S',inplace=True)
```

```
df.drop(['PassengerId','Name','Ticket','Cabin'],axis=1,inplace=True)
```

Categorical data to Numeric data

```
df['Sex'] = df['Sex'].map({'male':0,'female':1})
df['Embarked'] = df['Embarked'].map({'S':0,'C':1,'Q':2})
print(df)
```

```
     Survived  Pclass  Sex   Age  SibSp  Parch     Fare  Embarked  Family Size
1           1       1    1  38.0      1      0  71.2833         1            1
3           1       1    1  35.0      1      0  53.1000         0            1
6           0       1    0  54.0      0      0  51.8625         0            0
10          1       3    1   4.0      1      1  16.7000         0            2
11          1       1    1  58.0      0      0  26.5500         0            0
..        ...     ...  ...   ...    ...    ...      ...       ...          ...
871         1       1    1  47.0      1      1  52.5542         0            2
872         0       1    0  33.0      0      0   5.0000         0            0
879         1       1    1  56.0      0      1  83.1583         1            1
887         1       1    1  19.0      0      0  30.0000         0            0
889         1       1    0  26.0      0      0  30.0000         1            0

[202 rows x 9 columns]
```

```
x = df.drop('Survived', axis= 1)
y = df['Survived']
```

Splitting into testing and training set

```
x_train, x_test, y_train, y_test  = train_test_split(
    x,y, test_size = 0.2, random_state = 42
)
```

Training the model

```
model = LogisticRegression(max_iter=1000)
model.fit(x_train, y_train)
```

```
▾ LogisticRegression
LogisticRegression(max_iter=1000)
```

Testing the model

```
y_pred = model.predict(x_test)
print(y_pred)
```

```
[0 1 1 1 1 0 1 1 0 1 1 0 1 1 1 1 0 1 1 1 1 0 1 0 1 0 1 1 1 1 1 1 1 1 1 1 1
 0 1 1 1]
```

Accuracy

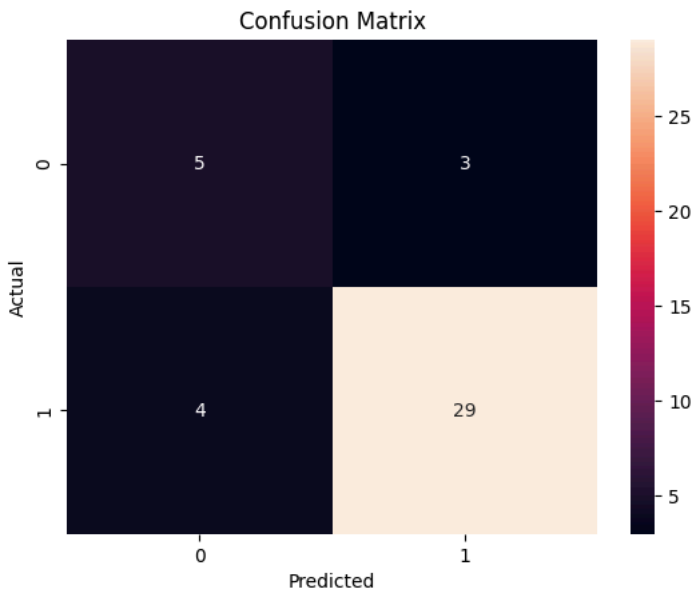
```
print("Accuracy", accuracy_score(y_test, y_pred))
```

```
Accuracy 0.8292682926829268
```

Confusion Matrix

```
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d')
```

```
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()
```



Start coding or generate with AI.